Anna Lo Piccolo

Ph.D.
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Brown University
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Education

Ph.D. in Earth, Environmental and Planetary Sciences 2021 – 2026
Brown University
Thesis: "Ocean Submesoscale Transport and Entrainment: Theory, Modeling, and Parameterization"
Advisor: Baylor Fox-Kemper

M.S. in Earth, Environmental and Planetary Sciences 2021 – 2023
Brown University
Advisors: Baylor Fox-Kemper and Christopher Horvat

M.S. in Physics of the Earth System 2017 – 2021
University of Bologna
Thesis: "Arctic Ocean Submesoscale Brine Driven Eddies: Modeling of a Sea Ice Edge Front"
Advisor: Nadia Pinardi
Co-Advisors: Christopher Horvat and Baylor Fox-Kemper
Grade: 110/110 *cum laude*

B.S. in Physics 2012 – 2016
University of Trento
Grade: 108/110

Research Visits

Visiting Research Fellow May 2023
Department of Physics, University of Auckland
Advisor: Christopher Horvat

Visiting Research Fellow Nov 2019 – Mar 2020
Department of Earth, Environmental and Planetary Sciences, Brown University
Advisors: Baylor Fox-Kemper and Christopher Horvat

Teaching and Outreach

Teaching Assistant Fellow Spring 2022
"EEPS1510 Introduction to Atmospheric Dynamics", DEEPS, Brown University
Undergraduate/graduate-level course

Co-supervision Spring 2025
Student Name: Spencer D. Francis
Primary Advisor: Baylor Fox-Kemper
Level: M.S. in Physics, Brown University

Outreach Activity 2023 – 2024
DEEPS CORES (Career Opportunities and Research in Earth Science), Brown University
At Hope High School, Providence

Software Languages and Ocean Models

Python, MATLAB, Fortran, C, LaTeX, MITgcm

Oceanographic Fieldwork

Conducted CTD-based dissolved oxygen surveys in Narragansett Bay, Summer 2025

Schools

- ECCO summer school 2025, Pacific Grove, California, May 2025
- Southern Ocean Summer School (SOSS) 2024, Cargèse, Corsica, May 2024
- Institute of Computing for Climate Science (ICCS) Summer School, University of Cambridge, Sept 2022
- Atmosphere-Ocean-Sea-Ice (AOI) winter school, Longyearbyen, Svalbard, May 2022

Invited Talks

- "Submesoscale Eddy-Driven Transport of Tracers: from Dynamics to Parameterization"
  Parcels Meeting lead by Erik van Sebille, Utrecht University, Dec 2025
- "Modeling submesoscale eddies in the wintertime Arctic Ocean"
  CMCC, Bologna, 2021

Conferences and Meetings

- Gordon Research Seminar (GRS) and Gordon Research Conference (GRC) on Ocean Mixing, June 2026
  Mount Holyoke College in South Hadley, Massachusetts, United States
  GRS oral presentation: "Vertical Transport by Submesoscale Eddies: a Parameterization for Entrainment"
  GRC poster presentation: "A Parameterization for Submesoscale Entrainment and Subduction of Buoyancy and Passive Tracers"
- International Liége Colloquium 2026 Submesoscale Processes in the Ocean, Liége, Belgium, May 2026
  Remote oral presentation: "Enhancing Entrainment in Submesoscale Fluxes: Parameterization Development and Impacts"
- Ocean Sciences Meeting (OSM) 2026, Glasgow, UK, Feb 2026
  Oral presentation: "A Parameterization for Submesoscale Entrainment and Subduction of Buoyancy and Passive Tracers"

- European Geophysical Union (EGU) General Assembly 2025, Vienna, Apr 2025
  Oral presentation: "Parameterizing Entrainment Induced by Submesoscale Eddies"
- CPT Annual Meeting 2024, Brown University, Providence, Aug 2024
  Oral presentation: "Parameterizing Entrainment and Subduction Induced by Submesoscale Processes"
- Ocean Sciences Meeting (OSM) 2024, New Orleans, Feb 2024
  Oral presentation: "Submesoscale Parameterizations Have Not Been Evaluated for Entrainment"
- SASIP (Scale-Aware Sea Ice Project) annual meeting, Bergen, June 2023
  Oral presentation: "Parameterizing Ocean Eddies at the Edge of Sea Ice Leads"
- SWOT AdAc Consortium Workshop, Brown University, Providence, Aug 2022
- SASIP (Scale-Aware Sea Ice Project) annual meeting, Grenoble, June 2022
  Oral presentation: "Eddies at Leads"
- Gordon Research Seminar (GRS) and Gordon Research Conference (GRC) on Ocean Mixing, June 2022
  Mount Holyoke College in South Hadley, Massachusetts, United States
  Poster presentation: "Energetics and Transfer of Submesoscale Brine Driven Eddies at a Sea Ice Edge"
- Ocean Sciences Meeting (OSM) 2022, remote, Feb-Mar 2022
  Poster presentation: "Energetics and Transfer of Submesoscale Brine Driven Eddies at a Sea Ice Edge"

Peer Review Service

Reviewer for *Journal of Physical Oceanography*, *Nature Communication*, *Geophysical Research Letters*

Administrative Experience

Student Representative at University of Bologna — 2017 – 2019

Extracurricular Activities

- MPOWIR WAVE Summit, Glasgow, UK — Feb 2026
  Women Advancing Vision and Excellence
- Unfiguring, Mahindra Humanities Center, Harvard University — March 2024
  Experiments in the practice of science and art, and interdisciplinary graduate student conference.
  Projection of "Ocean Eddies: a Sensorial Experience", documentary by Anna Lo Piccolo, Jacopo D'Angelo (independent musician), Francesco Fusaro (independent curator and artist), Alex Gonzalez (independent artist), Elva Granados Escartin (independent artist).
- 2022 RI C-AIM Vis-a-thon, RISD, Providence, RI, USA — Spring 2022
  Arts-based program of science visualization.
  Realization of "The Majesty of Sea Ice", a textile artistic installation on the status of sea-ice in the Arctic Ocean.
- Volunteer at Geological Museum of Dolomites, Predazzo, Italy — Summer 2017

Publications

[1] Anna Lo Piccolo, Baylor Fox-Kemper, Genevieve J. Brett, Tomas Chor, and Jacob O. Wenegrat (2026). "A mixed-layer eddy parameterization for entrainment fluxes". In preparation

[2] Anna Lo Piccolo, Christopher Horvat, and Baylor Fox-Kemper (2024). "Energetics and Transfer of Submesoscale Brine-Driven Eddies at a Sea Ice Edge". In: *Journal of Physical Oceanography* 54.7, pp. 1489–1501. DOI: 10.1175/JPO-D-23-0147.1

[3] Samuel Brenner, Christopher Horvat, Paul Hall, Anna Lo Piccolo, Baylor Fox-Kemper, Stéphane Labbé, and Véronique Dansereau (2023). "Scale-Dependent Air-Sea Exchange in the Polar Oceans: Floe-Floe and Floe-Flow Coupling in the Generation of Ice-Ocean Boundary Layer Turbulence". In: *Geophysical Research Letters* 50.23. e2023GL105703 2023GL105703, e2023GL105703. DOI: https://doi.org/10.1029/2023GL105703